

Enhancing Security Protocol Using IAM Integrated with AI: A Scalable Framework for Website Security Scoring

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B.Sc. in Computer Science

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Declaration

It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
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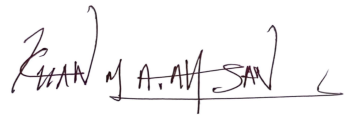
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Abstract

With more and more data in digital platforms, organizations find it harder to protect sensitive data. Identity and Access Management (IAM) allows for the only authorized users to use critical systems, but traditional IAM can't scale with the new breed of cyber-threats like credential theft, phishing and AI attacks. The research in this paper examines how artificial intelligence (AI) is able to make IAM smarter, better and more adaptive. In fact, AI-enabled Identity And Access Management uses machine learning, behavioral analytics and anomaly detection to discover active threats proactively in real time. In addition, it integrates XAI to improve interpretability, transparency, and compliance. This study examines the flaws found in standard IAM as we studied real word attacks like the SolarWinds and credential stuffing breaches, and we propose a scalable AI powered IAM framework. Results of AI driven IAM are tested and shown to decrease security risks in various domains (e.g. in finance, healthcare, defense) whilst improving compliance (GDPR, HIPAA) and UX. Based on this we conclude that the future of cybersecurity is AI integrated IAM — a proactive, intelligent, and scalable identity management solution.

Keywords: AI, IAM, Cybersecurity, XAI, Risk-Based Authentication, Defense in Depth, threat detection, continuous validation, multi-factor authentication, access control, biometrics and IAM systems, Deep Learning Authentication, Behavioral Biometrics, Role-based Access Control (RBAC), Cloud Application Security.

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Chapter 1

Introduction

1.1 Background

In the present digital world, organizations now heavily rely on digital platforms and interconnected systems in order to regulate their day-to-day operational activities. Different sectors including healthcare, banking, education etc depend on internet infrastructure to transform how they manage and store confidential information along with accessing and sharing data. In the light of emerging cyber threats, it has become a critical challenge to ensure secure access to digital assets for different organizations. Organizations have become susceptible to major risks through digital infrastructure dependency as cybercriminals purposefully take advantage of system vulnerabilities to gain access to confidential information.

Security practices developed during the years include firewalls as well as intrusion detection systems (IDS) and multi-factor authentication (MFA) which fight against cyber threats [25]. Identity and Access Management (IAM) stands as the fundamental solution which protects sensitive information because it enables organizations to handle digital identities and control system and data access. The IAM framework functions as a full solution to administer computer access for authorized users who require use of desired resources across defined platforms. IAM facilitates effective access through its established rules and processes and organized policies that defend credentials.

1.2 Rational Of the Study or Motivation

With the advancement of technology, the cyberattacks have also increased their strengths. The attack surface keeps expanding exponentially, proliferating the Internet of Things (IoT) meaning, it is more challenging to monitor and secure devices connected to the network [14]. On top of that, the adoption of cloud computing has driven the IAM processes even further complicated as organizations have to handle the identity and access control in the scattered cloud environments to comply with stringent security practices and regulations. As a result, the conventional IAM systems are proving to be less efficient against these much more complex cybersecurity threats. Attackers currently exploit enhanced methods like credential cracking along with social engineering phishing to bypass conventional defensive measures. credential theft where the passwords and usernames of users are stolen continues to be an extremely popular attack method experienced by organizations. Attacker access to

compromised credentials which grant them illegitimate entry into important systems so they can extract data and create disruptions or use stolen credentials to pretend to be valid users. The SolarWinds supply chain attack of 2020 remains one of the most widespread, destructive and complex cyberattacks in history that affected the U.S. government and corporations through massive breach of confidential data [12].

In order to overcome these growing obstacles, IAM systems need to be improved in order to accommodate real-time threats and minimize risks in advance. Here, Artificial Intelligence (AI) steps forward to be a game-changing solution. This is because incorporating AI into IAM in addition to the basic IAM functionality results in an increase in ability to forecast, identify, and prevent security threats. Machine learning based AI driven IAM systems use machine learning algorithms to analyze, automate and learn from patterns, detect anomalies and respond to threats ‘proactively’ ahead of time. Real-time access controls with these systems can identify unusual login behaviors, flag suspicious access attempts, and even in real time automate workflows to revoke access or isolate compromised accounts. For example, Microsoft Active Directory, a prevalent IAM tech, made it AI-powered so as to get stronger in standing off credential theft and unauthorized access [1]. AI algorithms can identify AI login patterns and investigate signs of a potential brute force or a credential stuffing attempt before a user in fact gets access to the system. Just as, AI can be leveraged to build out existing frameworks like Structured Threat Information eXpression (STIX) taxonomy to categorize and predict possible attack scenarios, to let organisations be one step ahead of the attacker [27].

While the implementation of AI results in security system enhancement opportunities, it presents multiple obstacles for IAM systems as well. The reason for this is, AI can also be used to develop more sophisticated cyber threats, for instance, AI generated phishing campaigns and malware. The necessity to advance IAM approaches persistently remains critical because organizations need to combat continuously advancing cyber adversaries. Much like other IAM applications the deployment of AI triggers data protection and regulatory compliance issues comprising GDPR among other standards. Progressive IAM solutions employ AI but organizations need to find equilibrium between security excellence and maintaining user confidence through clear disclosure and monitored system operation.

In conclusion, the involvement of AI within IAM systems is a significant change from the way businesses think of security problems. Such integration of the flexibility and the intelligence of artificial intelligence (AI) with the basic principle of information access management (IAM) may help businesses strengthen their ability to protect confidential data and ensure seamless access to resources. AI powered IAM solutions will create a protected digital environment which reduces dangers and enhances operational efficiency as the security environment develops. This thesis presents a full method exists to integrate AI-based enhancements into IAM infrastructure which addresses the current security problems eventually lead to better protected access control systems which operate more securely system.

1.3 Problem Statement

The traditional management approach for IAM systems fails to create complex solutions because of expanding digital dependence on IoT and Cloud Computing technologies. The combination of technical scalability issues and strict design architecture along with insufficient user security threat management creates primary security causes. Secure use of IAM suffers from inefficiencies as standard threat detection protocols from the past do not provide enough protection from contemporary cyber attacks against critical infrastructure. Artificial intelligence systems today struggle to connect with rigid legacy systems that in turn produce operational problems due to missing control measures that lead to regulatory challenges and ethical dilemmas. Failures create far-reaching devastating impacts. Security violations and operational failures stemming from such results align to produce financial consequences by forming a combination of legal repercussions and organizational reputation damage. Security professionals face unbearable levels of pressure from unjustified detection signals because they must shift resources from real security threats to respond to potential threats. The implementation of IAM systems demands time-responsive adaptability and scalable architecture combined with effective security provisions for ensuring complete data privacy and protection to solve these noted problems. A system built with Integrated Artificial Intelligence components in Identity and Access Management platforms shows resistance while increasing future security capabilities and automated responses that handle currently occurring cyber attacks to defend critical information.

1.4 Objective

The conjunction of IAM systems with AI technology resolves basic IAM systems' main operational challenges regarding massive user management and slow false alarm recognition together with adaptation problems. AI technology allows systems transformation through automated access control decisions which eliminate human-generated workload by implementing automated systems. Modern IoT and cloud computing infrastructures require this vital integration to provide real-time management of substantial user groups coupled with substantial device numbers during massive data streams transmission. AI Technology safeguards enterprise assets through fast-deployed protection services that merge with speed-based security systems and solutions for compliance needs. This research will focus on:

- The research initiative targets reactive anomaly detection methods as well as authentication systems that raise security effectiveness through fewer unnecessary detection alarms.
- A framework has been developed to address system integration problems which enable the move from traditional Identity and Access Management to AI-based solutions.
- Every operation within IAM benefits from Explainable AI (XAI) which provides transparency to establish trust and fulfill all ethical and regulatory standards.

- Real-world evaluations across multiple industries are necessary for the framework since practical applicability assessments and reliability tests must be performed across different industrial application environments.

1.5 Methodology in Brief

The research methodology uses systematic procedures with repeated testing to integrate AI into IAM systems for enhancing security capabilities and practical implementation capabilities. Researchers start their research procedure by choosing their topic and evaluating multiple academic publications at this initial stage. Organizations use analysis to discover their critical problems along with weaknesses in their present IAM system installations. The collected data allows for strategy development to analyze AI model datasets and provides training data preparation in support to identify appropriate AI capabilities for IAM requirements.

In the Implementation Phase the design and development of AI-IAM interface integration for authentication processes and access control and anomaly detection occurs. The system faces complete evaluation through testing mechanisms which require performance tracking alongside user analysis and security assessment. After the integration phase there exists an optional looping phase for improvement. Security guidelines compliance testing takes place during the ending phase while the solution expands to support large-scale operations and interfaces with the current IAM infrastructure. This part of the system maintains simplified complexity while minimizing operational costs to achieve security requirements. Real-world testing serves to validate the AI-based IAM framework when it operates in real-life environments to ensure both adaptation capability under further modifications and deployment readiness.

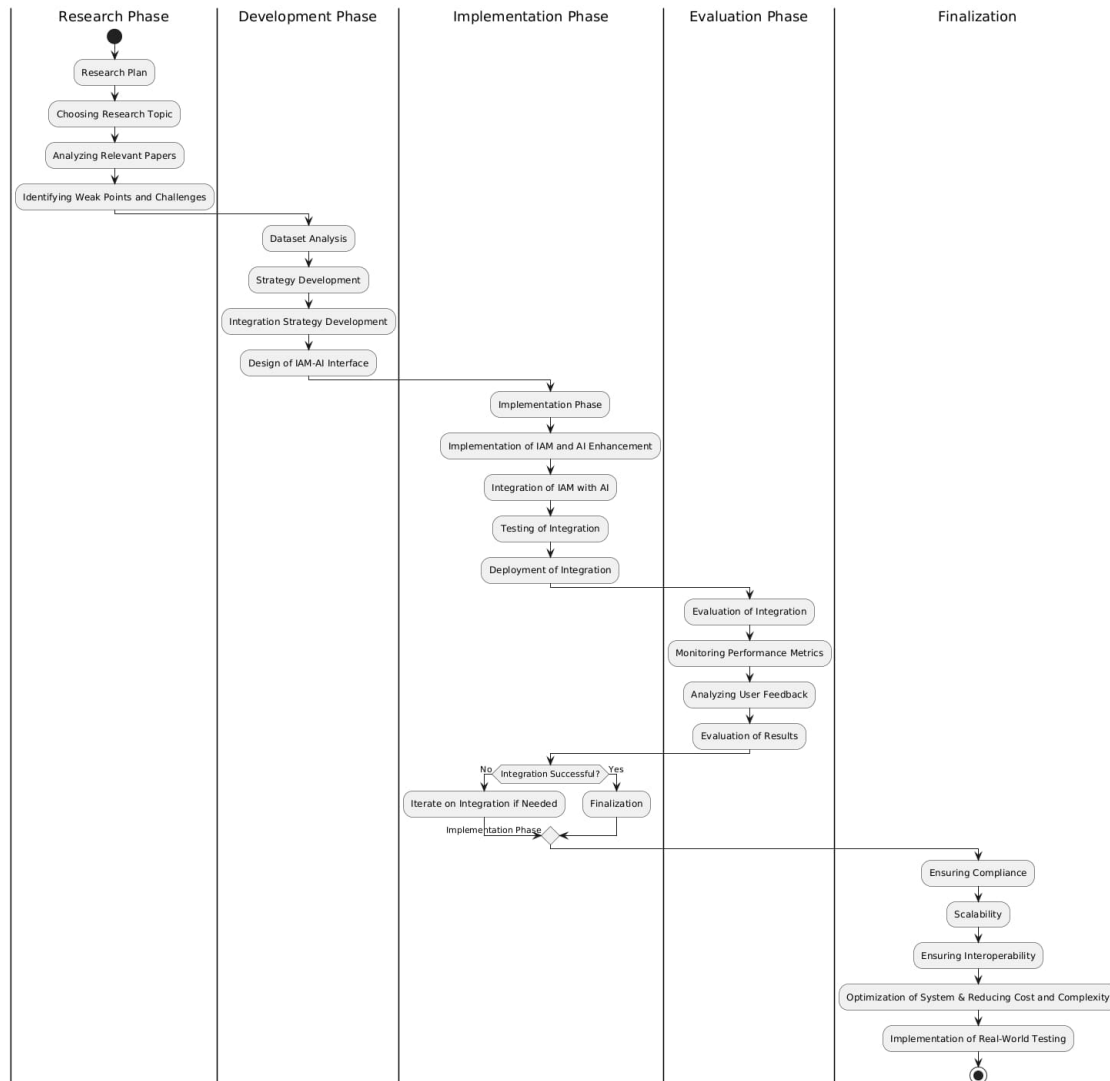


Figure 1.1: Workflow Acitvity Diagram

1.6 Scopes and Challenges

Scopes

1. **Security Improvement:** One of the key IAM functions can be obviously improved by AI through detecting anomalous events, blocking unauthorized access and fortifying authentication mechanisms.
2. **Automate the Identity and Access Management (IAM) process:** AI-powered IAM can analyze user behavior to grant or revoke permissions without human intervention.
3. **Threat detection and response:** AI integrations help to monitor for suspicious activity, identify anomalies, and respond / mitigate to threats even faster.
4. **Enhanced User Experience:** With the help of AI, strong security can be provided with also making the authentication process simpler (for example, adaptive authentication).

5. Scalability of IAM Systems: AI-based IAM system can scale easily and efficiently enable both enterprise and cloud IAM system
6. Regulatory & Compliance Adherence: AI-enabled IAM supports organizations in adhering to compliance standards through the automation of audit trails and risk management.
7. Cost Efficiency: AI-driven IAM solutions can help optimize operational costs by reducing manual tasks in identity verification and fraud detection.
8. Interoperability with Legacy Systems: The research can analyse the systems to ensure AI will bolt-on legacy IAM solutions without stopping operational bells.
9. Deploying AI-Powered IAM in Production: Conducting live deployments of AI-powered IAM systems to evaluate effectiveness, accuracy, and security enhancements.

Challenges

1. AI models that tackle IAM need access to high quality datasets and diverse training material which sometimes proves unavailable to the system.
2. The application of AI for access control systems becomes discriminatory whenever training is based on particular data that represents only partial system faults.
3. The AI system used for methodology implementations is susceptible to attacks that could negatively impact IAM system security levels.
4. AI introduces complexity because it fails to work with current mainframe technologies.
5. Client performance decreases when using AI-based IAM solutions which hinders the achievement of fast and secure system operations.
6. The transparency of AI decisions within IAM systems enables administrators to monitor system operations and solves problems more efficiently while maintaining proper accountability.
7. The implementation of AI-based IAM requires organizations to satisfy world-wide security regulations particularly GDPR and HIPAA and NIST through extensive legal examination procedures.
8. Users and organizations fail to adopt AI-driven IAM solutions because these solutions require validation through training before adoption can occur.
9. Installing AI-enabled IAM typically demands superior technological systems along with trained specialists who cost organizations high implementation expenses. The implementation of practical applications poses execution difficulties because real-world conditions need secure environments without security holes.

Chapter 2

Literature Review

2.1 Preliminaries

The concept of Identity and Access Management (IAM) has come into limelight from the beginning days of centralized computing when the user would depend for access on the password-based authentication only. In the 1990s, the statical one way method could not anymore live up to expectation since gradually the cyberattackers attained control over the access to the sensitive information easily through different means like brute-force attacks and leaking of passwords through multiple guesses. The authentication process became necessary to develop to mitigate phishing and leakages of information. Examining this, many revolutionary frameworks were utilized by the late 1990s and 2000s. One of the starting was with Role-Based Access

Control (RBAC) where a certain role was allocated to work upon that was required to input to the system to get access. This could not serve the purpose of solving the previous issues of cyberattacks either. The inadequacy later on introduced the then generation with better security solutions working with multi-factor authentication (MFA) and the token-based authentication in the early 2000s which came useful for checking the assurance of the user's identity using more than one means of approval terms. With the digital emergence of cloud computing structured for distributed resources, the pattern of biometrics and extra layer of protection policies itself struggled to keep up with the infrastructure changes and yet started coming to the control of hackers. This resulted in a cloud-native solution for IAM to accommodate the dynamic environments following which brought a new perspective to computer vision with the breakthrough of Artificial Intelligence (AI) and Machine Learning (ML) around the 2010s. The effect of AI has gained a massive global

attention since then. Gradually, AI controlled IAM technologies took over previous rule-based workstyle in handling all credentials using behavioural based control and detecting anomaly in the initial phase. In the meantime, dependency on Zero Trust Architecture (ZTA) made an impact in dealing with the unusual number of cyberthreats verifying every single type of access application. It faced few issues for not being able to cope with the legacy systems and the access process became monotonous and lengthy for the same users being verified to the system frequently. The risk factors and attacking possibilities were put under consideration to get used to the behaviour and how to act accordingly the taxonomy was established. And

while any of the issues arise, they wanted to put on real time action by adjusting the security measuring the levels of abnormalities which was basically the strategy of the Risk-Based Authentication (RBA) process. The application of biometrics

started flourishing from the year of 2018 which offered many identical detection processes with allowing individual's facial features scanning to fingerprint scanning abilities. This biometrics has remained prevalent to rely on against spoofing as a popular approach till now. Alongside, the appearance of blockchain technology had an interesting turn point in the workflow of IAM industries which provided a unique verification enabling method using cryptographic keys. The best part of the Blockchain IAM was it wasn't concise to any centralized lineup control, rather was distributed to several authorities to manage properly building better trust and be more reliable for the organizations. This system allowed the owner to reduce dependencies and work on the basis of Self-Sovereign Identity (SSI) but faced scalability problems for comparatively slower occurrence of workflow with transaction methods like Bitcoin, Ethereum etc. As a solution to this, an integration to the SSI systems with the blockchain standards was proposed to give an improved scalable experience by using blockchains Ethereum and Hyperledger Indy. The smart contracts gave growth to smart vehicles. The advancement in the technology fields can only

be grasped with data habituation by the AI-driven implementation. The feedback provided from the experiencing growth from the traditional IAM field to the modern structure can illuminate the ideologists with better understanding and it was always proven how integration of AI can be powerful to capture any kind of malfunctions right away. AI, blockchain and combinational learning continue on growing bigger, having us believe in the stronger resilience to the IAM functionalities and creating more faithful digital identities holding generation for the future.

2.2 Review of existing research

A research conducted by Olabanji et al. [24] investigates how AI is applied to IAM solutions for cloud infrastructure to improve scalability and address both threat detection boundaries and system inflexibility issues. Cloud environments experience scalability problems since their traditional IAM systems struggle to scale due to their fixed policies plus segmented infrastructure designs and human-controlled oversight systems. A combined analysis method uses structured surveys along with quantitative and qualitative survey data from 582 cybersecurity professionals including regression analysis to create research findings. The proposed system introduced AI through the combination of behavioral monitoring together with anomaly identification and deep reinforcement learning that received feedback. The implemented security measure enhances system capabilities while allowing it to adapt in flexible ways. This research highlights fundamental IAM problems which develop because users from different population groups struggle to accept IT systems and traditional rule-based authorization approaches and complex system environments create operational difficulties. AI-driven deployment has reduced negative anomaly false positive rates by 30% and optimized hardware has delivered substantial system reliability improvements. Before there existed single process based authentication methods that have evolved into a unified authentication method and system which balances

human-computer interaction. The work solved various demographic-related problems that previous research fields did not take into account.

Shaik et al. [15] investigates AI technologies that leverage Zero Trust Architecture for boosting IAM qualities to enhance security models through setting up context-aware systems. The previous traditional models lacked the capability to handle changes in the risk-driven environment properly. The system was designed through machine learning to establish User Behaviour Analytics (UBA) models targeting application competence for detecting correct user actions that would allow efficient work without system interferences. The ZTA integrated an ongoing learning system to adapt to this strategy which decreased unnecessary interruptions to 40% at the same time enhancing its ability to detect abnormalities. The implementation of adaptive authentication functions as a major solution tool that fixes convenience issues and resolves previous work-based gaps. The framework established in this research brings an effective AI-based solution for Zero Trust identity and access management where it addresses typical security challenges and adaptation problems. The value of these compound strategies with ethical considerations for modern cyber security domains can improve through practical development enhancements.

Vitla [26] discusses in his publication that AI enhances IAM systems by including anomaly detection capability with adaptive learning characteristics and scalability potential along with regulatory compliance features. Current organizations struggle to execute effective security solutions since traditional IAM systems operate by combining manual provisioning methods with written regulatory codes. The research shows that deep learning operates in real-time protection of system-level attacks by utilizing adaptive authentication methods with behavioral biometrics. Predictive analytics systems must acquire user access requirements for upcoming periods ahead of compliance audits that review automated access provision generated from affinity-based policies. A dual approach of encryption policies that secure user data links with monitoring systems for dynamic policy enforcement enables both customer trust and regulatory compliance. The AI system performs automated provisioning work that covers 75% of tasks while decreasing false positive identification by 40% through its developmental advances. Security policies' flexible framework enables the system to modify operations with live behavioral data thus creating a better experience while boosting system performance. The regulatory testing method for automated systems needs minimal user input when addressing defects from earlier systems. Security and scalability integration takes place through system components in this research conducted by the research team. The system demonstrates superiority to traditional RBAC models by integrating predictive analytics and behavioral biometrics features that enable it to boost real-time choices through learning procedures. The system framework complements privacy-secured techniques with scalable systems to create operational procedures which boost work process efficiency.

Neelakrishnan [23] analyzes how Artificial Intelligence can boost proactive security measures for data access in cloud applications. Traditional systems encounter recurring problems since the expansion of cloud based working and monitoring limitations to the activity of users given detecting harmful intentions. With working upon analyzing the characteristics of anomalies, a User Entity Behavior Analytics (UEBA)

system was established to act on any kind of suspicious or unusual activities being detected. Before the system gets messier to maintain, some profound ideas were demonstrated which includes: Being more productive in preventing rather than giving alarming signals, Decentralizing the official controls making the system more reliable and in compromising to take over-guard measures to bring better user benefits. The architecture was sustained with four planes: naming activity feeder plane, aggregator plane, analytics engine place and action driver plane. The activity feeder plane collects information in raw form from the end users and gives it to the system for further analysis. It primarily gathers log files and login attempts using tools like Azure AD and Okta. Then the aggregator plane processes that raw information that has been collected after filtration and builds a user-entity relation structure form. The analytics engine plane is a core part of the architecture which functions with many AI-driven algorithms to situate the problem basics and patterns. Later the generated risk score points allow the workflow in analyzing how to mitigate the anomalies in the system by producing the policy recommendations. Finally, the action driver plane is mainly the platform of taking action to the insights it allocated. This plane supports SOAR system integration as a remedy to the triggers it faced. Organizations achieve real-time scaling along with proactive prevention through this system implementation. The platform develops initial algorithms during a 90-day analysis period which depends on historical data and identifies boost detection accuracies. The study fills in essential gaps currently present between current cloud-native data protection systems in the market. The solution identifies network-based fraud automatically in real time rather than waiting until regular systems perform later analyses while simultaneously stopping insider threats through preventing credential theft along with other types of data exfiltration activities. An intelligence system that distributes analytics through multiple operational areas constitutes this decentralized framework. The addition of risk measures from validated entities working on advanced logging procedures enables operational forensic analysis through these integrated measures. The system implements a mechanism which enables administrators to track suspicious activities although they retain control of transitioning automated tasks to human decisions. This latest system presents improved capabilities over previous technologies through its alert fatigue elimination and rapid incident response functions and its capability for tracking suspicious user behavior patterns. This method separates itself from traditional UEBA technologies because it uses interactive AI-human feedback systems in addition to static analytic technology and contextual information. The system enables thorough criminal and human-based user-entity mapping that enhances combination between investigation processes and real-time threat detection features. The decentralized microservice architecture of this system enables rapid business expansion by solving scalability problems which affect single-vendor UEBA deployments.

Yaacoub et al. [18] explores significant amounts of data to be ingested from the increasing dissemination of IoT Devices that often contain vulnerable user data. Participants can obtain decentralized access to train AI model data stored on their local devices to keep the data privacy-preserving using Federated Learning (FL). FL systems have identified vulnerabilities along with several challenges when integrated with IoT components which allow attackers to perform model inversion and membership inference methods. The research analyzes the FL and IoT relationship

by investigating security risks while presenting solutions and suggesting developmental avenues. A thorough examination of FL algorithms and protocols operates across IoT settings by separating solutions based on their cryptographic properties (homomorphic encryption and multi-party computation) from those without crypto functionality (differential privacy). The research methodology uses structured threat identification and mitigation strategy development to analyze IoT-specific challenges and device heterogeneity and resource constraints and communication overheads. Several contributions are made by the research, such as, it presents a detailed view of the various forms of attacks occurring on Federated Learning (FL), which describes their way of targeting servers and clients hampering the custodianship of the users. Then, this research makes a comparison to the protective measures in cryptographic and non-cryptographic processes and talks about their progressions with reviewing. This ensures the enrichment of knowledge resulting in future benefits to protect FL systems. Furthermore, this work proposes some necessary workstyles to boost the security of FL systems fighting to cope up in some environments like Internet of Things (IoT) or in other frequently changeable environments having no capability to contain vast resources. Moreover, it gives a clear viewpoint about the ways of handling network models in different types of network environments. The previous studies that this paper refers to had a gap in not explaining about interconnected systems and there was no discussion about considering the challenges that might come through from the IoT related environments of lacking storage incapacities and uneven data distribution. This was solved in this presented study by offering efficient security ideas for IoT settings. Besides, the research also established practical connections between FL theory with competing in the real world scenarios, ensuring that there remains a profound scalability to the system.

Among other attributes, ZTA provides a holistic solution to cybersecurity challenges as presented by Muhammad et al [7]. The ZTA model presents the 'Never Trust, Always Verify' principle, which states that no user or device can be fundamentally trusted under this model. Serious security concerns can, however, be alleviated through continuous authentication, authorization and encryption. In the layered security model, or defense-in-depth, multiple security layers are implemented: physical security (biometric authentication), network security (firewalls and intrusion detection systems), application security (secure coding and web application frameworks), host security (endpoint protection), data security (encryption and access controls), and identity management (enforced authentication mechanisms), which will allow other layers to be completely unaffected elsewhere even if single ones do not succeed independently. The combination of these layered security frameworks and international cyber security standards improves the organization's overall security resilience, compliance, and scalability. Among the layers, an NIST framework provides risk assessment and continuous monitoring, while ISO/IEC 27001 implements an Information Security Management System (ISMS) dealing with risk mitigation. Finally, OWASP identifies web-based security vulnerabilities. When such security measures merge, an organization can reduce its attack surface while creating a stronger data security vertical along with global rules aimed at developing an adaptive cybersecurity system.

Singh et al. [16] discuss Identity and Access Management (IAM) as the cornerstone

of security in the quest for safe user verification and permission management. IAM implements procedural and technology-related management to tackle digital identity management and control sensitive information access through policy frameworks. Recent IAM solutions work with SSO and MFA, including cloud-based identity management platforms such as Azure ATP, KeyCloak, and CyberArk, to eliminate this vulnerability against unauthorized access, promote user identity safety and to prevent identity breach. The paper lays focus on a growing issue regarding password protection, as weak combinations of passwords are chosen by users and hence they form a higher risk against digital security attacks. Tools like SailPoint and Okta that have centralized control and automated access control can solve those concerns. AI technology can play a crucial role in significantly enhancing IAM in anomaly detection, enabling it automating security policy management. AI-based IAM solutions strengthen user behavior examination while tracking unusual system interactions which allows for immediate dynamic access control capabilities. Research evaluates cloud environment IAM implementation through investigation of hybrid IAM models which seek security equilibrium with user experience. The major security benefits of IAM must overcome two fundamental obstacles which include implementation challenges and problems with role management and insufficient planning. In short, to secure digital assets, streamline user authentication and ensure regulatory compliance in this high-tech era organizations need to implement robust IAM frameworks.

[9] According to Victoria Adenola's finding, digital resource access control together with identity authentication and authorization functions through IAM establish its core position in cybersecurity systems. The traditional authentication systems reveal their deficiencies because people make too many mistakes in data protection and because the authentication methods have weak points and malicious employees exist. The application of AI to IAM technology delivers an active solution that enhances access privilege control through better monitoring operations alongside administrative functions and enforcement protocols. Real-time anomaly detection as well as behavioral analytics and biometric verification serve as key risk mitigation capabilities that AI provides for security systems. IAM systems supported by machine learning techniques discover unauthorized access while generating fewer false positives to improve interface user experiences through adapting to new security patterns. Medical investigations reveal how AI enhances identity and access management (IAM) systems by combining IBM Security Verify with Oracle Adaptive Access Manager (OAAM) as well as Cisco Identity Services Engine (ISE). Unifying AI technology with detection solutions provides both user authentication help and prevents unauthorized entry to systems. The beneficial features of AI-powered IAM systems come at a cost which includes user privacy vulnerabilities along with ethical problems as well as security risks to AI model functionality. The complete exploitation of security access systems through AI capabilities depends on improving algorithms and controlling ethical concerns and regulatory compliance.

The paper by Mohammed [5] analyzes how artificial intelligence (AI) transforms identity and access management (IAM) through precise and efficient automated user access control procedures. It shows core functionality of IAM and demonstrates AI-based security solutions provide enhanced real-time access decision secu-

rity by delivering context-based analysis. By using machine learning, IAM detects peculiar user activities which subsequently enhances authentication protocols and lowers manual error frequencies. The research details AI-based identity verification methods that combine biometric data and behavioral analysis to build efficient security frameworks that no longer need conventional passwords. It investigates how IBM applies AI to develop IAM innovations that link adaptive access rules with fraud monitoring systems for maximizing enterprise protection. Additionally, it demonstrated how Artificial Intelligence systems advance regulatory requirements and strengthen risk oversight functions and minimize management expenses involved in traditional access process operations. IAM technology development will establish complete automation systems enabled by AI-driven behavioral access controls to replace conventional authentication methods and result in cybersecurity changes through enhanced operational efficiency as well as accuracy and adaptability.

The manuscript by Ghadge, N. [19] delivers an advanced framework to perform authentication with the integration of machine learning. The combination of AI and behavioral analysis components in IAM enabled machine learning technologies for real-time anomaly detection and risk-based authentication. Basic experiments were applied and for real-time monitoring and taking actions based on the attacks, Security Information and Event Management (SIEM) integration was pulled which gathered events from diverse controls. It identifies if the system is restricting the user from multiple login attempts and notifies to check the suspicious activity and gives an automated response accordingly. It basically appeared a lot useful in terms of strictly restricting malicious activity from occurring. This research mainly worked on adjusting with the behaviour and improves the overall process giving a unified view. However, during the application of the framework itself, there were some limitations, as bringing up a dry test if not applied thoroughly could produce vulnerabilities. This is considered as it's Weakness, SIEM cannot accommodate huge flow of inputs. The User Behaviour Analytics (UBA) are quite a smart way to deal with tracking possible fraud and displacement of any kind of detecting inner-level threats that may come along. Moreover, incorporating Security Orchestration, Automation, and Response (SOAR) can provide better visibility, and cloud-based SIEM solutions can help with any dynamic impacts and visibility at a lower price. The integration of IAM with SIEM will help in implementing the AI and ML processes, which can help in smoothing the process between automatic approvals. Based on this approach would be focusing on lightweight yet scalable machine learning models focusing on improving automated response mechanisms, following conditional and preventative measures, and coupling IAM with cloud-based solutions for hybrid workforces. To summarize, organizations can significantly improve their cybersecurity resilience by combining proactive threat detection mechanisms with real-time monitoring. As threats continue to evolve, combining SIEM with AI-driven IAM solutions can offer better protection for sensitive data and organizational assets.

With the advent of nature in technology constantly changing, the risk factors and attacking possibilities were put under consideration to get used to the behaviour and how to act accordingly the taxonomy was established. And while any of the issues arise, they wanted to put on real time action by adjusting the security measuring the levels of abnormalities. [13] Pöhn, D., & Hommel, W. structured the Taxonomy

for Identity Management related Attacks (TaxIdMA) with characterized forms of attacks taking professional interviews and learning from other reliable sources. Attacks coming for system level, user level and in the technological fields constitute the major categories of attacks. It focused that an ease method can be demonstrated with good understanding about how the possible attacks can be supplemented across the layers of protecting the credentials. To give an extra layer of protection, the frameworks were established as Structured Threat Information Expression (STIX) taxonomy which will enable one to keep one step ahead of the attackers any time before the cyber criminals come for attacking. This is basically a language form of expressing risk based commands to the prompt which was structured from testing in various application forms in domains like Internet of Things (IoT) and self sovereign identities (SSI). The weakness in the article goes again in coping with the newly introduced attack vectors everyday.

The use of biometrics gained momentum in around 2018 and evolved many identity detection processes, such as scanning facial features using facial recognition technology or making use of the fingerprint as biometric identity. This biometrics has remained prevalent to rely on against spoofing as a popular approach till now. Simultaneously, the penetration of blockchain technology brought a considerable change in Identity and Access Management (IAM), verifying users and systems through cryptographic keys. In contrast to traditional IAM solutions that are based on central authority, the best part of the Blockchain IAM was it wasn't concise to any centralized lineup control, rather was distributed to several authorities to manage properly building better trust and be more reliable for the organizations. This system allowed the owner to reduce dependencies and work on the basis of Self-Sovereign Identity (SSI) but faced scalability problems for comparatively slower occurrence of workflow with transaction methods like Bitcoin, Ethereum etc. Naghmouchi, M., Ayed, H. K. B., & Laurent, M. [8] proposed an integration to the SSI systems with the blockchain standards to give an improved scalable experience by using blockchains Ethereum and Hyperledger Indy as a solution to this. . In this system, we leverage the implementation of smart contracts such as Decentralized Identifiers (DID), Verifiable Credentials (VC), Decentralized Public Key Infrastructure (DPKI), and DID Authentication Protocols for identity verification and improving the security of smart vehicles. Nonetheless, challenges persist, including the need to ensure seamless interoperability between different blockchain networks and enhancing the usability of SSI solutions for non-technical users. The future research involves optimizing the blockchain interoperation as well as extending these IAM solutions beyond smart vehicles to a larger range of IoT applications.

Traditional IAM technologies being prone to inefficiencies caught the attention regarding the necessity of bringing a transformation to relying on deep learning technologies more. The invention of Self-Sovereign Identity (SSI) had been used prominently for a long time being efficient in reducing the risks in user privacy concerns. It checks on a structured testing of individual information and protects digital identities with cryptographic attestations, obviating the necessity of givers of credentials in a centralized way. Glöckler, J., Sedlmeir, J., Frank, M., & Fridgen, G. [11] highlighted the issues with a blockchain based SSI prototype that leverages AI for securing identity management. The methodology it used was a design based multi-

phased Design Science Research (DSR) providing a detailed description of reviews termed as Systematic Literature Review (SLR) using across enterprises. For the improvement of the ongoing prototype, the SSI based IAM prototype was made with Hyperledger Aries Cloud Agent in Python (ACA-Py) which was deployed to the real world implementations. It has a few number of limitation factors that could not be neglected even after providing a well developed decentralized management system. One of those is that there can arise a blockage to adopting the interoperability with legacy IAM solutions which can hamper in the progress of integrating components coming to no use later on. Besides, usability concerns come upon having to let non-technical users. This will be needed in the future with an increasing number of testing and exploration for IoT devices and customer identification methods.

Khare & Arora [20] in their research paper looks to explore how artificial intelligence (AI) and machine learning (ML) can modify the risk based identity verification processes. The main point of this paper is that AI/ML technologies can greatly amplify the accuracy, productivity, and adaptability of identity verification by detecting fraudulent conduct and performing real time risk assessment, as well as providing advanced authentication methods to improve the veracity of the user experience. A survey based research method was used with a meta analysis approach in this study utilizing data sourced from AI and ML experts and also cybersecurity professionals, to evaluate the effectiveness and challenges of AI and ML in identity verification. The researchers assessed response to how well algorithms like supervised learning and deep learning perform at tasks like anomaly detection, fraud prevention and biometric authentication. The accuracy and adaptability of the AI models, challenges in deployment (i.e., false positives, security), and the influencing factors with risk scoring (for instance, transaction history and behavioural pattern) were key variables. The survey did have value, but it heavily relied on subjective input from a small sample size and it could never be sent out ‘live’ to test. Responses to data analysis were classified into agreement levels, identified trends, and areas for improvement. While there has been emphasis on the technical and ethical aspects of AI driven IAM, the study does not provide empirical validation or a set of sector specific application insights that limit its wider scope of applicability.

[3]In the paper titled ‘The Interaction Between Artificial Intelligence and Identity & Access Management: An Empirical study’, Mohammed proposes that integrating artificial intelligence (AI) with Identity and Access Management (IAM) can substantially advance cybersecurity and simplify user authentication systems while improving identity management. According to it, not only can we solve traditional IAM limitations such as inefficiencies in access control, lack of scalability and vulnerability to sophisticated cyber threats, but also increase the quality of common user interactions. The research employs a literature review based methodology which is mainly a qualitative and analytical approach that reviews existing literature on traditional IAM practices and the integration of AI technologies, such as machine learning, behavior analysis, and risk based authentication. The study looks at using AI for automation, anomaly detection and adaptive decision making in IAM. Furthermore, it stresses the application of AI for real time monitoring, predictive analytics and contextual access management. The authors propose a conceptual model that discusses AI and IAM to bring a more secure, efficient and adaptive

system. It points out how AI can be used for automatic authentication of low risk cases, for identification and detection of anomalies of user behavior and for the improvement of regulatory compliance. According to the study, by transforming IAM to a proactive system through AI, it becomes possible to achieve better operational efficiency and security.

Mohammed [2] investigates how IAM (Identity Access Management Systems) can be enhanced through the application of intelligent authentication. The paper mainly focuses on password management and multi factor authentication within the domain of IAM inorder to improve its usability and effectiveness. The author assesses the drawbacks of conventional IAM processes and suggests unified intelligent authentication as a way to maximize operational effectiveness, enhance security, and simplify user administration. The author mainly relies on literature review of existing research and concepts of IAM to identify the shortcomings in traditional IAM practices such as point solutions, static passwords and through comparative analysis, dives deeper into more advanced approaches such as multi factor authentication and AI-driven systems for improved security, scalability and efficiency. The author highlights the need for leveraging the power of AI into risk based authentication and behavioural analytics in order to streamline the user authentication process through reducing vulnerabilities in password related tasks and ultimately improve security compliance.

In the paper “Analysis on the Role of Artificial Intelligence and Identity and Access Management”, Muppa [22] evaluates how Artificial Intelligence (AI) technologies help Identity and Access Management (IAM) systems achieve better cybersecurity results. Identity and Access Management ensures appropriate data protection yet its current systems deal with managing multiple identities and preventing unauthorized access. This research investigates progressive authentication concepts alongside analysis of technology variables together with user populations and system interface elements which affect IAM function. AI-powered Identity and Access Management (IAM) system development aims to deliver practical design recommendations for building efficient usable access solutions.the methodologies that were used are coupled quantitative data analysis with qualitative observations to provide a thorough insight.for data collection the researchers distributed 700 surveys to cybersecurity professionals.they received 582 valid responses.The survey explored subjects connected to IAM user experience together with technology specifics and the functions of artificial intelligence within this domain. Multiple regression analysis of the collected data revealed connections between systemic variables which included both hardware and software configuration data along with system performance measurement.for Security solutions utilizing facial recognition boundaries and fingerprint identification methods as biometric verification techniques, multifactor and risk-based authentication. For cloud based systems AIM systems operated with Artificial Intelligence solutions.the security features proposed are Encryption to protect data transfers and both Privileged Access Management (PAM) and Role-Based Access Control (RBAC) operate as access management systems defined for targeted operational roles.Functional and technical system optimization plays a critical role in enhancing AI performed IAM operations.This research demonstrated how user demographic characteristics determine trust levels and uptake of AI practices

within IAM systems. The need for privacy combined with transparency measures was emphasized because both elements enhance user confidence. My proposal included continuing learning algorithms to boost system operational efficiency and adaptive capabilities. Standard technologies should receive standardized quality requirements to decrease performance inconsistencies on different platforms. Multiple studies now confirm the potential of Artificial Intelligence to perform automated system enhancements for Identity and Access Management systems. Supports findings on the importance of encryption and multifactor authentication in cybersecurity. This research extends previous work through its analysis of demographic diversity effects and privacy concerns on user acceptance behaviors. The study presents thorough statistics indicating how different hardware/software system settings influence AI operational performance. Survey data obtained from self-reporting subjects can produce faulty biased results. The study focuses solely on cybersecurity experts therefore it overlooks broader general user groups. IAM systems based on cloud models overlook fundamental difficulties which predominate in both on-premises and hybrid infrastructures. AI biometric authentication requires maintaining complete user privacy. Solutions must deal with the diverse hardware software compatibility issues within different organizations.

The paper by Arora and Tewari [10] examines how Zero Trust Architecture (ZTA) when integrated with Identity and Access Management (IAM) and Artificial Intelligence (AI) improves cybersecurity capabilities. As an alternative to perimeter-based defensive approaches ZTA migrates towards verifying every element while suspending trust at their core. This architecture establishes ongoing verification alongside stringent access rules while executing real-time dynamic assessment mechanics across all system elements. ZTA reaches peak threat detection by deploying AI to conduct real-time identification of security threats while enabling adaptive security solutions and preserving high compliance protocols. The research examines how ZTA applies its security features across cloud systems and IoT devices and remote operations and microservice deployments. The research analyzed previous research connecting ZTA and IAM with AI to find both gaps and potential solutions. It defined main ZTA components by establishing the Policy Engine (PE) alongside Policy Administrator (PA) and Policy Enforcement Point (PEP). The implementation which was proposed by the researchers uses role-based access control (RBAC) together with machine learning (ML) models for developing adaptive authentication systems. The zero Trust principles are continuous verification of user and device identities and Least privilege access control. The system implements real-time threat detection functions through behavioral analytics capabilities furthermore, user and device behavior acts as a basis for security policy adjustments. To integrate IAM centralized identity management and intelligent authentication using biometric and multifactor techniques are necessary. ZTA principles became activated to establish security practices that overcome traditional perimeter security models and AI technology features in the paper due to its ability to provide live threat discovery alongside dynamic security protocols. So a proposed merge of ZTA with IAM would provide user-friendly administrative capabilities maintaining regulatory adherence and blocking unapproved system access. It showed ZTA successfully defends cloud systems and IoT networks while also protecting microservices platforms and remote worker connections. The proposal also offered guidelines to deploy ZTA concepts into enterprise infrastructure while pri-

oritizing machine learning solutions for permanent system adjustments. This paper builds upon previous ZTA theoretical studies by merging AI systems into practical real-time implementation. Dynamic authentication together with AI analytics helps the research identify the complete scope of Identity and Access Management. The experiment results verify former research by showing how ZTA blocks servers from establishing unauthorized connection routes and prevents illegal session transitions. The research extends the foundation of assessments which focus on trust within ZTA systems from previous research. ZTA together with AI systems depend on advanced programming expertise and strong hardware infrastructure to function through their complex framework. The use of ZTA becomes more intricate during operations involving a combination of cloud infrastructure since the implementation of scaling processes remains challenging.

According to Tamraparani [17], modern IAM systems face massive challenges having management issues due to working with large amounts of datasets and constantly changing information, making it difficult to detect the malfunctions and fraud patterns attacking the system. Research demonstrates how AI machine learning detection along with anomaly detection and pattern recognition systems effectively identify and stop fraudulent actions. The paper works to achieve advanced fraud detection functionalities through AI technology while resolving existing issues related to data quality together with false positive errors and system scalability requirements. Historical fraud data sets provided the foundation for developing machine learning (ML) models which detect future fraudulent activities. The research combined filter and wrapper approaches to improve prediction accuracy levels in datasets that contained imbalances. The ML models proposed in this paper are The Related Feature Subset Model (merges extraction and selection methods to yield the best results for fraud detection procedures.) and the researchers also implemented supervised learning algorithms including Logistic Regression together with GRU (Gated Recurrent Units) and Isolation Forest to detect anomalies. The proposed model operated using vast datasets from Kaggle among others while trials were conducted to compare outcomes beyond FDAM-LSTM conventional methods. This technology assists in two main applications: finding odd patterns, classifying financial transactions, and detecting discrepancies. The accepted algorithms consisted of RF-CFD (Related Feature Subset Model) and Enhanced Logistic Regression (LFS-ELR). The approach researchers used included a combination of dimensionality reduction techniques with correlation-based filters for processing and enhancing datasets. The model performance assessment used Accuracy and Precision and Recall and F1 Score with ROC curve evaluations. Our approach combined filter and wrapper methods for feature selection in order to improve fraud detection efficiency. An RF-CFD model framework was developed by the authors to enhance the detection accuracy of fraud in big datasets. Proof was provided showing how anomaly detection with GRU yielded superior results when used instead of traditional LSTM approaches. A method was developed to resolve the issues of unbalanced datasets alongside a real-time rapid fraud detection framework for IAM systems. The team engineered a framework using Machine Learning and scalability features which works for enterprise datasets and applications. The model achieved better detection accuracy by merging feature selection approaches with GRU units which represented normal application behavior. The proposed research analyzed extensive unbalanced datasets although other

studies primarily worked with smaller more balanced datasets. Research established previously shows that ML stands vital in fraud detection therefore this work builds additional insights from past studies. Research introduces enhanced feature selection approaches along with modern model optimization strategies which improve existing fraud detection frameworks.

2.3 Summary of Key Findings

AI-powered Identity and Access Management (IAM) is revolutionizing cybersecurity by improving authentication, anomaly detection, and access control. AI-driven IAM systems lower false positives in threat detection (by 30–40%), automate user provisioning, and adjust to user behavior, enhancing both security and operational efficiency. The combination of AI and Zero Trust Security Architectures (ZTA) enhances real-time risk-based user verification, and federated learning facilitates distributed security in IoT settings. Moreover, AI improves fraud detection, compliance assessments, and user behavior analysis, yet issues such as algorithmic bias, privacy worries, and integration with outdated systems persist.

A significant trend is the transition from fixed rule-based IAM to flexible, AI-powered access management, guaranteeing scalability and responsiveness to changing cyber threats. Nevertheless, AI models need transparency (Explainable AI) to foster user confidence and avoid bias. Ethical issues like AI-driven surveillance and privacy threats underscore the necessity for technologies that protect privacy. The lackings from the practical approaches and long term research will result in incapability to handle an increasing amount of resources reducing the proficiency. It is required that for better future outcomes, AI should be standardized in a more scalable way along with developing security solutions for hybrid clouds should be considered hereby restructuring AI-based IAM systems in an ethical and reliable way for the users.

Chapter 4

Design and Development

4.1 Design Process or Methodology Overview

4.1.1 Data Collection

The dataset titled “Cloud Access Control Parameter Management” (October, 2024) has been obtained from the Kaggle platform in the format of a publicly shared csv file in order to use for this research [21]. Kaggle is a well known public platform where data is curated for various machine learning and data science practices. The dataset used for our thesis contains numerous features which provides an overview about the access control settings, policies and control strategies applied in various cloud services. Thus, the dataset is coherent with IAM, which is the theme of this research.

The original data set comprises 100,000 data points and 51 features. These features are associated with cloud-based IAM configurations and security practices. It has some significant columns that directly correlate with Identity and Access Management (IAM) which consequently play an important role in deciding on the security status of an organization. The main IAM-specific features are Authentication Mechanisms (including MFA, SSO, and password-based login) that define the way identities are verified securely. Authorization Models, such as the RBAC and ABAC identify the approach to assigning access rights to users depending on their roles or attributes[6]. User Identity Management, User Roles, and Access Levels are the core features that define the level of individual access and authorization in any organization. The Access Revocation, Privileged Access Management, and Least Privilege Principle guarantee that the access is provided correctly and is removed when it is not required, as it minimizes risk. Session Management, Third-Party integrations, Federated Identity, Audit Trails are other great features that work collectively to facilitate secure, traceable and policy oriented IAM activities.

Additionally, the dataset includes the target variable Security Score for each configuration/row that reflects the overall security strength following IAM strategies. The target column has the following classes (2,3,4,5) where higher values of score represent stronger security measures and vice versa. All together, these features provide a strong basis for analysing the strength of IAM policies adopted in any multidimensional organization.



Figure 4.1: Methodology Flowchart

4.1.2 Data preprocessing and Analysis

Data preprocessing is an important step in any research, especially crucial in case of security related tasks to ensure the data integrity for overall performance of machine learning models. An overview of the strategies adopted in order to make the dataset more suitable for model training has been given below:

Feature Encoding

Among the features, 7 columns include object type variables (categorical/textual) and the rest contain boolean type variables(True/False). In order to apply machine learning algorithms to categorical data, the most important features Authentication Mechanisms, Authorization Models, Access Levels, User Roles, Cloud Service

Provider, Geolocation Restrictions, and Data Sensitivity Classification were encoded by using Label Encoding. In this approach each unique category is masked with a unique integer so that the data can be interpreted by the models without rendering itself to high dimension. For example, in the Authentication Mechanisms column the values Password, MFA, SSO and Biotic were coded as 3, 2, 1 and 0 respectively. Correspondingly, the categories of User Roles were associated with the values 0, 1, 2, and 3 for Admin, Developer, Guest, and Manager respectively. This part of data pre-processing is especially important for computability and interpretability of different categories of any feature inside a machine learning model.

Boolean Conversion and Cleaning:

After an Initial inspection of the dataset, it was found that numerous features contained boolean values (True/False). These were systematically converted to binary values (1 for True and 0 for False) using the map function in order to facilitate numerical computations. This process was essential in bringing the dataset to consistency. It enabled downstream algorithms like the tree based classifiers, support vector machine, etc to process the input variables efficiently preventing any type errors or any inconsistency.

Label Filtering and Data reduction:

Security Score entries that did not receive a value of 3 or 4 were merged to increase the effectiveness and clarity of the modeling process. This decision was taken after it was noted that extremely high/low values (e.g., low score of 2 or high score of 5) were infrequent and were likely to add skew to the analysis. The inclusion of scores 3 and 4 enables more balanced comparisons between the middle to high security categories without having issues linked to the presence of outliers and imbalance of classes.

Feature Engineering:

In order to enable a more comprehensible interpretation of IAM security status, the raw numerical scores of the Security Score (2, 3, 4, 5) were translated into broader qualitative risk types. This conversion was done using three engineered features - **Smart Risk Score**, **True Count**, and **Risk Admin No MFA** which are derived from the existing feature set. True count is the total number of activated security features denoted by positive boolean value, Risk Admin No MFA denotes users not having multifactor authentication (highest security) in spite of having higher privilege over the system and lastly Smart Risk Score is a feature developed using a custom risk scoring formula that penalizes critical features not being enabled. This approach did not only improve the capacity of the model to identify the underlying risk factors but also opened the concept of domain-specific intelligence in the model.

Handling Class Imbalance:

When analyzing the IAM dataset, the class imbalance in the Security Score variable was recognized as a crucial issue. Although the most infrequent classes (class 2 and 5) were dropped, the skewness of the target variable persisted. To mitigate this

a normalization approach aimed at reducing the possible bias was adopted which included clustering the label class 4 further into 3 groups (40, 41, 42) using K-means clustering and later reclassifying the security score values into three new groups: low, medium, and high. The objective for this conversion was to attain an overall balanced distribution of the target variable. This strategy promotes fairness and effectiveness of supervised learning models by avoiding dominance of one particular class.

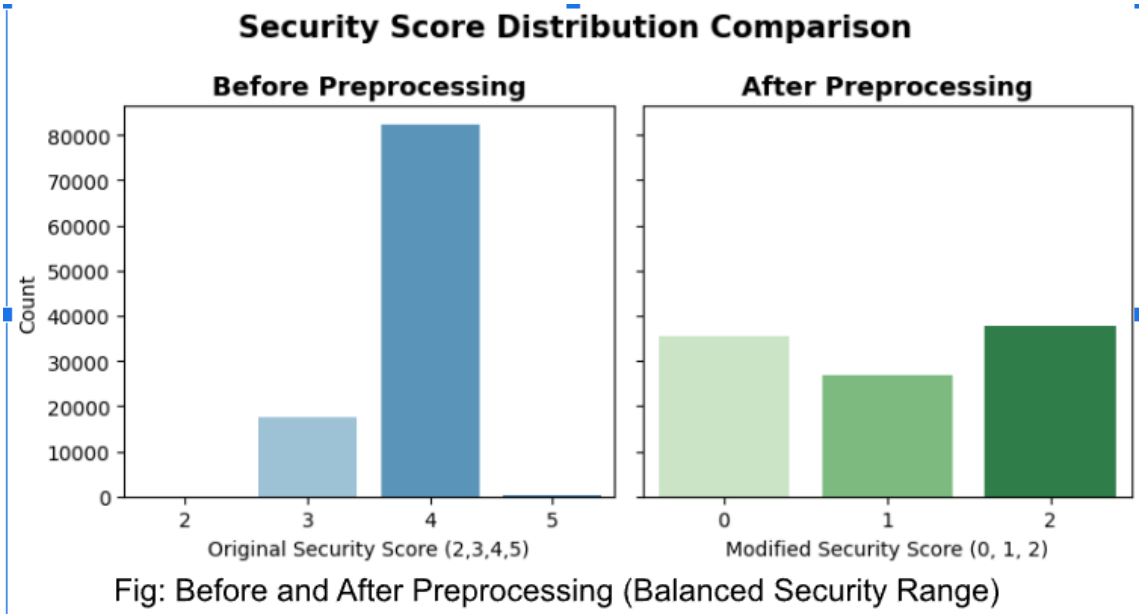


Figure 4.2: Security Score Class Distribution

The algorithm recurrently scanned the range of bounding possibilities with lower and upper thresholds and found the combination, which resulted in minimum variance of the classes. In particular, it was assumed that observations smaller than the lower threshold (smart risk score $i = 14$) were low, those between the thresholds medium, and those above the upper threshold (smart risk score $i, 16$) high. This re-categorization trained the initial numeric target to produce a well-balanced categorical variable that fit into the multi-class classification. The cleaned data with all the re-categorized labels were saved as a CSV file to use during the next training and testing phases of the model.

4.2 Preliminary Design or Design (Model) Specification

4.2.1 Overview

This section presents the implementation of multiple machine learning models designed to evaluate the security strength of a system based on its IAM (Identity and Access Management) configuration. The goal is to predict the Security Score categorized as low, medium, or high in order to assess how vulnerable or protected a given policy setup is. After preprocessing the dataset and encoding categorical features, we trained and evaluated multiple classifiers, storing both performance metrics and

confusion matrices. Among all the trained models, the best performing ones have been discussed.

4.2.2 Model Architectures and Training

XGBoost

XGBoost (Extreme Gradient Boosting) model is an excellent ensemble model which computes and accumulates a series of decision trees in order to minimise errors most efficiently. This model was used in classifying security scores, and the aim was to use its effective gradient boosting structure to deal with complex decision boundaries as well as feature interactions in data associated with IAM such as this one. XGBoost also works very well on structured categorical data such as the access control dataset. Having encoded categorical features and mapped the labels of classes, the model was trained on the balanced dataset and demonstrated an impressive accuracy on the test set of 89%. It also got a high result in all three classes resulting in an overall F1-score of 0.887. As demonstrated by the confusion matrix, Xgboost had perfect classification of all three classes - low, medium and high. In general, XGBoost did a fine job processing the resampled, encoded data to provide a well generalized model which can reasonably predict the level of vulnerability of any system.

LightGBM

LightGBM is a fast scalable library that implements gradient boosting. It is faster than the traditional boosting techniques because of its leaf-wise tree development and its use of histogram-based techniques and competitive accuracy. Because this model processes categorical data and can train fast on large datasets, this model was specifically appropriate to our IAM dataset. It registered an F1- score of 0.809. LightGBM, whose training set was resampled and label encoded where the class imbalance problem was addressed through feature engineering and clustering, achieved an overall test set accuracy of 83%. The model showed perfect classification of the three classes similar to the performance of Xgboost . All in all, LightGBM was a successful multi-class classifier in the application of IAM-based security scoring, where it provided an optimal balance between the accuracy and the computational time requirements.

Catboost

CatBoost, a gradient boosting algorithm, is natively capable of handling categorical features, which makes it a suitable algorithm to use with IAM datasets that contain such variables as User Roles, Authentication Mechanisms, Cloud Service Provider etc. CatBoost was selected in this study due to the capability to analyze complex relationships in structured data with the minimal preprocessing. The accuracy (0.986) and F1 score (0.985) of this model was very good and demonstrated a higher quality of classification. The confusion matrix demonstrated that the precision of the result was high in each category such that there were 6,980, 5,227, and 7,505 correct predictions in Low-risk, Medium-risk, and High-risk respectively. Few Medium-risk records had been falsely labeled proving that the model is capable of discerning small shifts in access control settings. The combination of the good computational

qualities coupled with its interpretability entitles it to security-oriented applications in an IAM system.

Keras Neural Network

The neural network has been trained by Keras as a multi-layer perceptron that evaluated the quality of modeling complex feature interactions in IAM through deep learning. It did the best overall and its F1-score 0.992. Normalized IAM characteristics and one-hot-encoded security score were used to train the model with two hidden layers (128 and 64 neurons respectively) and dropout to regularize the model. It also registered near ideal learning depending on the confusion table with a 0.992 accuracy, precision and recall. The fact that validation and training loss were decreasing throughout the 15 epochs demonstrated that there were no poor generalization characteristics, and there were no signs of overfitting. The number of errors was rather small (e.g. 115 misclassifications class 0, 28 class 1 and 9 class 2). This is a great indication that the model did a great performance in obtaining both linear and non-linear dependency in the IAM feature, a fact which makes the model extremely viable and applicable in real-life systems, where the systems are expected to be accurate in every possible way.

QDA

The Quadratic Discriminant Analysis is a kind of probabilistic guiding model which makes an assumption of normal distributing features in each class. It works quite well in interpreting from a separated variance structure based on Bayes theorem. For the training model to have class-specific matrices of low, medium, high risk results, QDA model was assumed to capture the non-linear boundaries well. QDA has overpowered many more of our data train models with 0.8 F1 score. Our model train test results show that it could be valuable for insights, though the data per class actions in further imbalances, there was a risk for it to fail miserably.

Performance Comparison: Top 5 IAM Risk Scoring Models

Model	Train Accuracy	Test Accuracy	Precision	Recall	F1-Score
Keras Neural Net	0.9998	0.9992	1.0	1.0	1.0
CatBoost	0.9994	0.986	0.98	0.99	0.99
XGBoost	0.9754	0.8955	0.89	0.89	0.89
LightGBM	0.8743	0.828	0.82	0.81	0.81
QDA	0.817	0.809	0.8	0.799	0.8

Figure 4.3: Performance Comparison

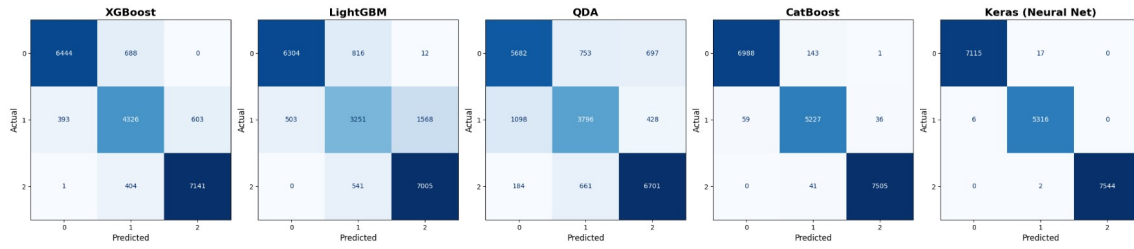


Figure 4.4: Confusion Matrix

4.2.3 Model Insights and Framework Proposal

The preliminary establishment of the IAM assessment tool was fundamentalized into an Machine Learning integrated Django-based web design framework trained from XGBoost chosen to be the perfect suited model.

Model Analysis and Specification:

Apart from the multiclass classification report aimed at identifying weak spots, a varied group of models- from traditional approach to deep learning train and tests were performed to prepare it for the real time environment. It aimed at minimizing data overfitting, which could be a common challenge in cybersecurity logs. Random forest, Extra Trees, and K-Nearest Neighbours, despite getting a full trained score, the test accuracy dropped by over 0.25. This proved the underperforming tendency by these models when fed with any unseen IAM setup.

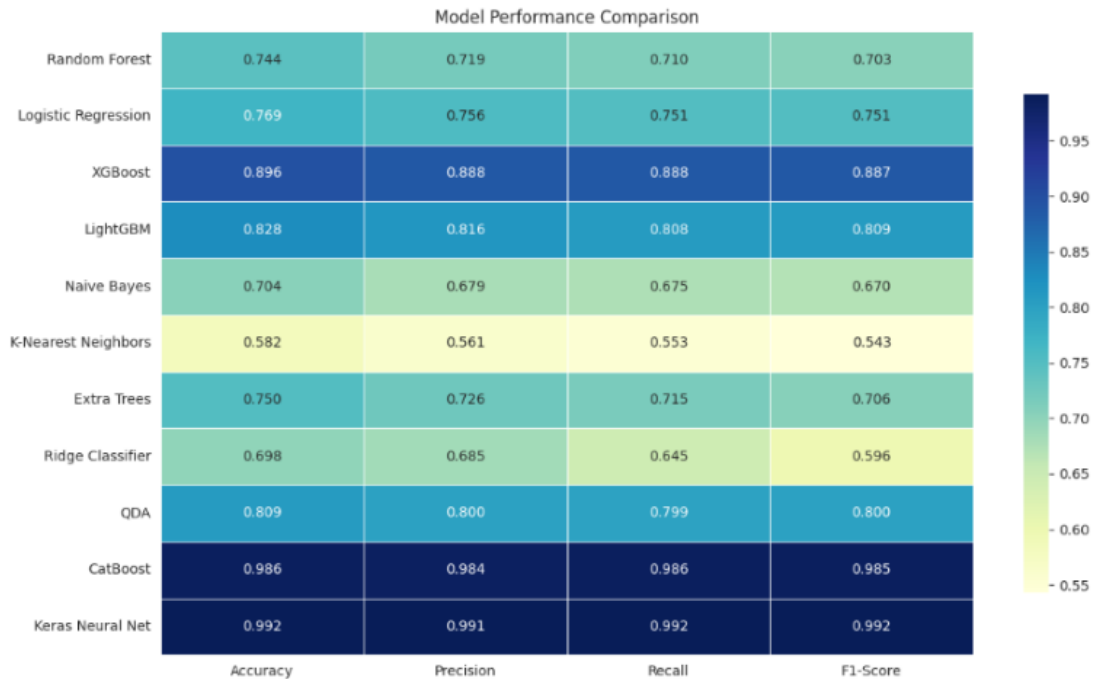


Figure 4.5: Model Performance

On the other hand, XGBoost and CatBoost outperformed with satisfactory results on both train and test nature. XGBoost gave a minimal train-test variance scoring

89.55% on test. The working principle of extensive hyperparameter tuning strategy and faster approach of working on large datasets with L1 (alpha) and L2 (lambda) results in fulfilling high-dimensional security detection related purposes precisely. The potential risk scores were identified with its high resistance to noise functionality making it resilient to complex workloads. CatBoost even outperformed all other comparative analysis, showing excellence in 98.60% test perfection, closer to matching the network baseline, while not further diving into deep learning. The handling technique of the CatBoost model did best with minimal preprocessing reading the categorical and boolean features.

However, in order to deploy to the website, the requirement was to simplify the complexity overall which XGBoost delivered a faster training time in the preprocessed dataset. The lightweight web deployment acknowledges more explainability to serve in responding to compliances. XGBoost Classifier (iam security model.joblib) and encoders (iam encoders.joblib) was loaded to the model after training process enabling the assess function to collect from user and web inputs and generating recommendation techniques (e.g. enabling MFA for admins). The well driven balance between generalising and handling technicalities serves the iterative assessment steps in the Django application plan, shaped by considering XGBoost as the best alignment as a real time visionary tool. [4]

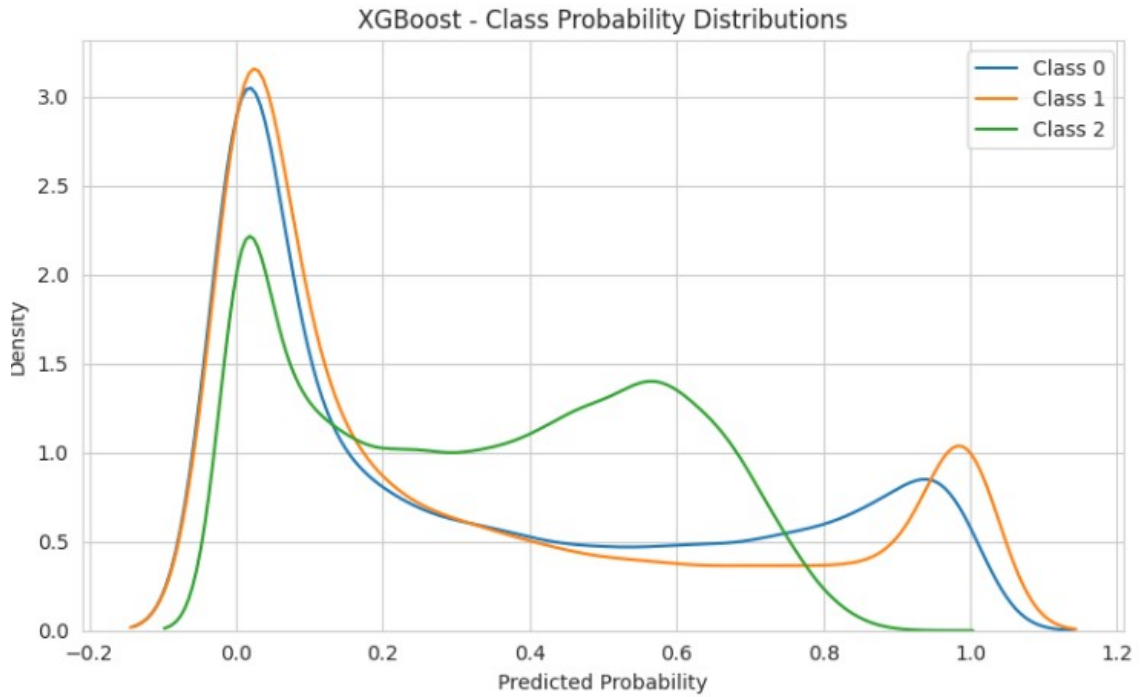


Figure 4.6: Class Probability Distribution: Xgboost

Web Application Architecture :

The robust Model-View-Template (MVT) Pattern for the Django web application framework offered modularity with QuickAssessmentForm allowing user insights validation. Later, the user inputs were merged with default policies, calling assess IAM policy, and outputs detailed result reports based on internal mechanisms as trained

towards. It strategizes and defines endpoints with `urls(py)`, with enriching modularity configuring the `ScannerConfig` based on the established framework.

The enlisted features gave a considerable amount of ideas about the IAM assessment tools feature invigilation. From the larger amount of necessary factors, around five to six critical recommended items were chosen to communicate with the users about their insights and rebuild a correlation view working as a default value in the backend. The efficiency enhancement takes a more better approach with layer based work procedures.

The XGBoost model was utilized as the first layer of working mechanism to give a correct base to the recommendation engine. The security standard based recommendation will not point the vulnerabilities straightly rather it will suggest the upgrading techniques. Based on the framework, A SLM (Short-Language-Model) will be built as a chatbot for the companies based on the Django framework soundation. The company security standard ideas can be known well enough to guide and reshape towards through the user friendly seamless and low risk model. The higher risk security scores will give a frontier knowledge to various systems and build a stronger safety to minimize cyber attacks.

The external sources will be readily available to the framework transformation using the label encoders, which will leverage feature importance metrics as due progress. A responsive interface was established to take inputs from the users via a dynamic process with modular designing to allow scalability to the users handling routing and persistent logging of the data smoothly. The design supports scalability integrating with different accessible factors through the use of `joblib` serialization, updating any changes without causing interruption to the application flow. Continuous learning and deployment agility techniques made to structure `models(py)` supervising Django ORM integration will pave the way to deal in evaluating significantly and audit trails to relation databases. With this, batch analysis support and practical usability would be ensured through the progress of the data driven, iterative configurations.

Limitations

Although the dataset is well defined in terms of access control and identity features, they have certain issues that affect our analysis and the performance of our model. For example, there was a severe imbalance in the categories included in the Security Score label as most of the values were in one of the classes (e.g., score 4). This bias affected the effectiveness of classification models and required resampling or relabeling approaches carefully to train in a balanced manner. There were also numerous features in the form of booleans (e.g. `True`, `False`) which had to be preprocessed (at great scale), encoded, and made consistent. Another challenge is that there are some

features whose values are uninterpretable. Fields such as "Authorization Models" or "Access Levels" lack standard definition or domain-specific mapping, which brings a challenge to know what the correct security implications of such values would be without assumptions. Furthermore, the dataset does not include contextual data such as user behavior, time-based access patterns, or historical attack logs, causing it to limit its ability to use it for real-time risk scoring. Finally, the evaluation of

association or dependencies among features is limited because of lack of metadata (e.g., feature source, timestamps, or correlation with real-world incidents), causing the models to rely on static, surface-level indicators. Improving these shortcomings would make the dataset a valuable asset to solid IAM threat modeling and security assessments using AI.

Chapter 5

Conclusion

Identity and Access Management (IAM) receives a paradigm shift from AI because it delivers better security and simplified procedures as well as improved operational effectiveness. Through AI-based identity and access management systems one can detect abnormal behavior and fraudulent activity with immediate results which increases cyber defense capabilities while decreasing security alerts. However this technology faces several difficulties which prevent its mass acceptance. One such difficulty is the use of biased algorithms within AI models which lead to problems with unfair access control making many ethical concerns necessary. AI-based behavioral analytics require strong protective solutions which include privacy-preserving AI and machine learning because of the privacy risks they present. When XAI transparency lacks quality it produces trust problems that make compliance with regulations more challenging. Implementing AI-driven IAM with existing legacy systems creates an essential barrier because it requires solving compatibility challenges and expensive deployment expenses. The pursuit of research should concentrate on ethical trials related to AI alongside framework standardization and real-world implementations of scalable solutions that advance security efficiency and trust between users for sustainable AI adoption in the realm of IAM.

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